



6.4 Concept 3: *Higher Education/Longhouse/Elders Buildings Solar and Storage Microgrid*

This concept would combine the Higher Education, Longhouse, and Elder’s buildings into a single solar and storage microgrid. This project would use a 45kW solar array on the roof of the Longhouse and medium-scale battery equipment near the existing utility transformer, with a new gas generator to support the three buildings for the design 7-day outage.

Although this was evaluated as a constructable system, it is not recommended for development due to the following:

- Site Constraints; buried water and sewer utilities surrounding the buildings to the South and East; Important gathering areas and natural view corridor on the North and West.
- Uncertainty around new facility construction location, and potential shading impacts.
- Longhouse roof structural concerns.
- Elders and Higher-Education building roofs poor choices for solar (North facing)

In winter, outage resiliency is determined primarily by the generator fuel available. In other seasons, however, when sufficient solar power is available to recharge the battery during the day a utility outage can last much longer and not require as frequent operation of the generator.

Smaller systems focusing on critical loads may also be feasible, however, the uncertainty around future development adjacent to the building makes the risk of solar shading and exterior battery locating of

concern. The Longhouse is the only of the three buildings that appeared to have unobtrusive areas to install batteries inside the building but rooftop solar is of concern as discussed above.

System Details

- Provide resilient power to 100% of building loads for a 7-day design outage beginning early January.
 - 31kW Peak demand
 - 412kWh/Day average
- Estimated Savings with Microgrid
 - Net Metering benefit of \$4,615 per year
 - No Demand charge savings
- Rooftop Solar array
 - 45kW-DC rooftop array
 - 34.2kW-AC rooftop inverter optimizer system
 - 45,470kWh/year production; 104% offset of Higher-Ed Meter
- Battery energy storage system (BESS)
 - 15kW Sol-Ark Hybrid Inverter (x4)
 - 153.6kWh Battery (HomerGrid units x4)
- Generator
 - 75kW Gas generator with 500-gallon LP tank.
- Electrical Use (Both Meters)
 - Annual Use: 119,547-kWh
 - Peak monthly use: 14,297-kWh in April
- Electrical System
 - 50kVA Padmount Transformer
 - 240/120V 200A Panels in each building

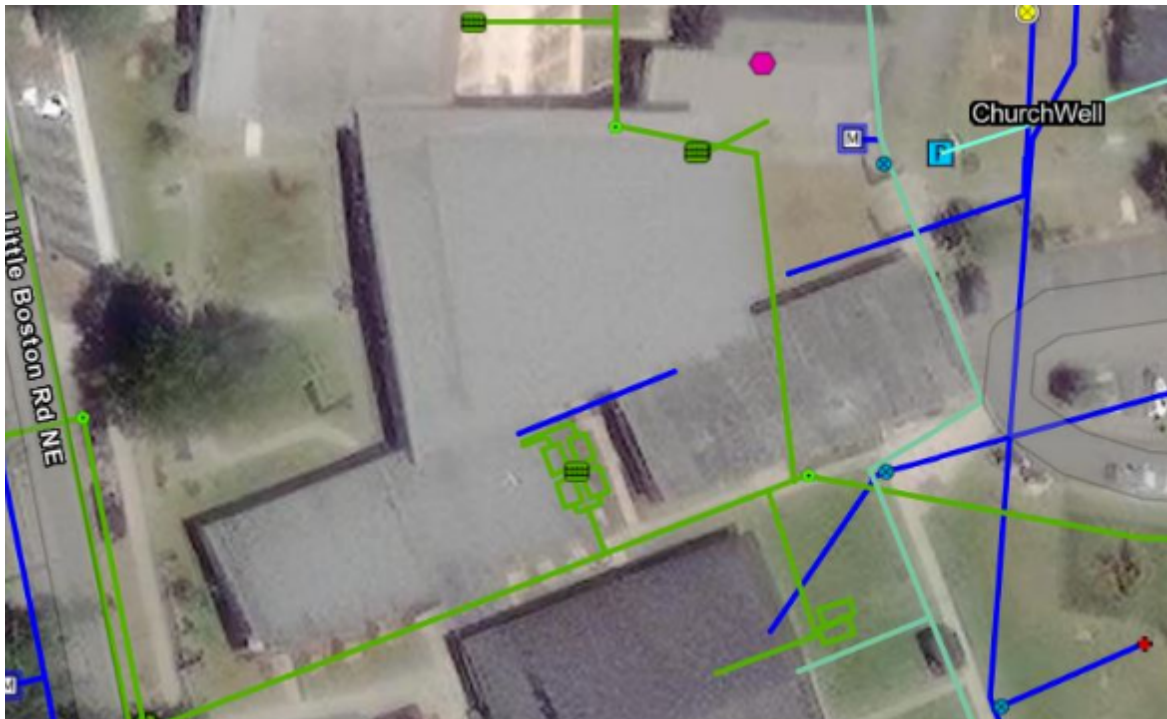


Figure 18 Concept 3: Buried water and sewer utilities around buildings⁷ These limit new equipment locations



Figure 19 Concept 3: Future development areas in orange and blue and promenade/view path constrain buildings on all four sides⁸

⁷ Image from Tribes Utility GIS database. Special Thanks to Mike Rorem.

⁸ Image from 2014 Master Planning document; 20-year Master Plan Site Plan